

Impacts of the SDWA Arsenic Rule on Arsenic Occurrence in U.S. Drinking Water [†]

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Abstract

This study provides the first retrospective analysis of the causal impacts of the 2001 US Safe Drinking Water Act (SDWA) Arsenic Rule on arsenic (As) occurrence in public drinking water systems. Using publicly available SDWA compliance data, we employ a difference-in-differences strategy that compares As concentrations before and after the promulgation of the Rule for treatment relative to control water systems, defining

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treatment based on whether a water system had ever sampled As concentrations above the Rule’s 10 micrograms per liter ($\mu\text{g}/\text{l}$) threshold *before* the Rule was published. We find that annual mean As concentrations in treatment systems fell by almost 5 $\mu\text{g}/\text{l}$ ($p < 0.01$) after 2001 relative to control systems, representing a decrease of about 38% compared to pre-treatment levels. In light of the Rule’s projected net costs at the time it was promulgated, assessment of the Rule’s realized impacts using actual measured As concentrations from a nationwide sample is an important first step in retrospective benefit-cost analysis.

Introduction

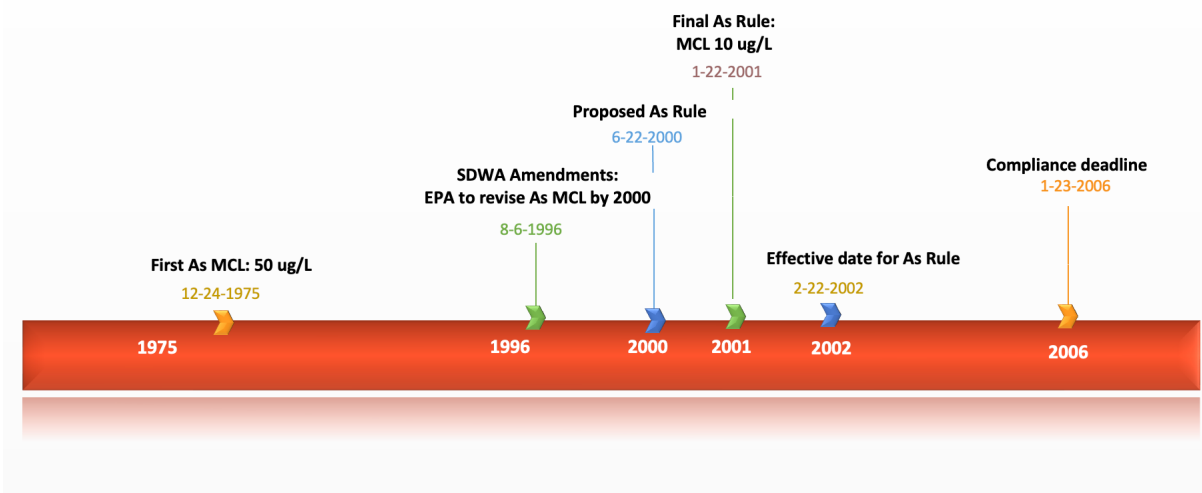
Arsenic (As) naturally occurs in the environment in soils, rocks, and volcanic materials. Some industrial and agricultural processes, for example those in glass and semiconductor manufacturing and the application of herbicides and pesticides¹, also contribute As to the environment. Inorganic As has implications for health: it is a known human carcinogen^{2–4}, and the literature documents associations between exposure to inorganic As and ischemic heart disease⁵, diabetes⁶, adverse pregnancy and birth outcomes⁷, and neurodevelopmental effects⁸. Due to these and other human health risks, the concentration of As in drinking water is regulated in the United States under the Safe Drinking Water Act (SDWA).

On January 22, 2001, the U.S. Environmental Protection Agency (EPA) promulgated the Arsenic Rule, which lowered the maximum contaminant level (MCL) for As from 50 to 10 micrograms per liter ($\mu\text{g}/\text{l}$), with compliance required by January 23, 2006.⁹ The Rule also implemented changes in compliance monitoring for As. Prior to the Rule, a system found in violation of the (weaker) 50 $\mu\text{g}/\text{l}$ standard was required to provide three additional samples within one month of the violation. After two consecutive samples below the 50 $\mu\text{g}/\text{l}$ MCL, the system returned to the default sampling frequency of at least one sample every three years for groundwater systems and annually for surface water systems. Under the Arsenic Rule, a violation of the 10 $\mu\text{g}/\text{l}$ MCL triggers quarterly monitoring. After a

minimum number of samples, the state regulator can determine when the violating system returns to the default sampling frequency.

The Arsenic Rule has been controversial, in part because its projected economic benefits were less than projected costs at the time it was promulgated in 2001¹⁰, and in part because of the potential for high compliance costs for systems serving small communities^{11,12}. Following the controversy, EPA postponed the effective date of the Rule by a year, to February 22, 2002, during which time EPA requested additional public comment and formed expert panels to reassess the new standard. Figure 1 presents a timeline of events associated with the Rule.

Figure 1: Arsenic (As) Rule Timeline



^a Note: Figure depicts a timeline of events relevant to the regulation of As in drinking water in the U.S. In 1975, the National Interim Primary Drinking Water Regulations established the first enforceable As standard of 50 $\mu\text{g}/\text{l}$. The SDWA Amendments of 1996 directed EPA to propose a new arsenic standard by January 1, 2000. The initial proposed Arsenic Rule considered a standard of 5 $\mu\text{g}/\text{l}$ but the final Rule established the new standard at 10 $\mu\text{g}/\text{l}$.

While the SDWA requires rigorous prospective benefit-cost analysis of new and revised MCLs, nothing in the statute requires retrospective review. Retrospective analysis is needed, however, to assess the causal impact of a policy on the response of regulated entities (in this case, public and private drinking water providers) once a Rule is passed, as well as the health impacts and other effects of that response^{13–15}. Since the Arsenic Rule’s passage, research has explored some of its potential effects. For example, Foster et al.¹⁶ show that violations of the As standard initially climbed between 2006, the first year compliance with the new 10

$\mu\text{g}/\text{l}$ standard was required, and 2008, and have declined steadily since then. They also show that most of the increased violations occurred in small systems and those using groundwater rather than surface water as a primary source. However, because the standard became more stringent, violations remain higher than they were prior to the Rule¹⁶, and it can be difficult to know how As occurrence and exposure have changed when considering changes in violations. Examining about 300 water systems in California, Grooms¹⁷ shows that As concentrations in drinking water may have declined as a result of the 2001 Rule, but that violations do not result in subsequent reductions in As concentrations among violating systems. The As Rule may also be associated with improved birth outcomes in California.¹⁸ In the epidemiological literature, Nigra et al.¹⁹ analyze urinary As concentrations in the United States, comparing levels in 2003-2004 (after the Rule was promulgated, but before compliance was required) with those in 2013-14. The authors show a decline in dimethylarsinate and total As concentrations between these two periods in nationally-representative samples (though of different individuals) among those who self-report access to public drinking water, but not among those using private water wells. While not a causal analysis, the estimated associations are consistent with the Arsenic Rule having lowered exposure to As in public drinking water in the United States.

Given controversy over the Arsenic Rule's benefits and costs, uncertainty over the Rule's impacts on intended outcomes, and the recent release of a final IRIS Toxicological Review of Inorganic Arsenic by the EPA¹, an important input to future rulemaking under the SDWA, additional work is needed to understand the causal impact of the Arsenic Rule on As occurrence and exposure. The literature evaluating the Arsenic Rule has been thin in part because historically the only available data for assessing regulatory impacts were violations data. The violations data identify when a water system exceeds an MCL, but not by how much. It is also impossible to observe any changes in As concentrations below the MCL in the violations data. Thus, violations are a noisy and incomplete measure of drinking water quality.

As an essential first step toward a retrospective quantification of the Arsenic Rule’s benefits, we use publicly-available data capturing spatial and intertemporal variation in As concentrations in drinking water to estimate the Rule’s impact on arsenic occurrence in community water systems. We assemble SDWA compliance samples from 1995 to 2019 using occurrence information collected for the Rule itself as well as three Six-Year Reviews of Drinking Water Standards by the EPA.^{20–23} An advantage of using compliance samples is that one can create a panel of As sampling records for systems before and after the Rule using *actual* As concentrations for regulated water systems as opposed to numbers of violations, which may mask important variation in As occurrence. The SDWA compliance samples are routinely collected by states and used for monitoring and enforcement, but they have only recently been employed by researchers.^{24–26}

Materials & Methods

Public drinking systems in the United States serve about 90% of the population, with a majority of people receiving drinking water from community water systems (CWS).^{27,28} CWS are public water systems that serve at least 25 people or at least 15 connections year-round. To estimate the impacts of the Arsenic Rule, we use SDWA compliance samples for CWSs, assembled from four sources: the occurrence analysis for the Rule itself, the Six Year Review 2, the Six Year Review 3, and the Six Year Review 4.^{20–23} These data include sample information from 24, 45, 51 (including Washington, DC) and 48 states over the periods 1980-1997, 1998-2005, 2006-2011, and 2012-2019, respectively. We focus on water systems consistently classified as CWS during this period because CWS were required to meet the new, more stringent standard, and because we observe As samples for many CWS both before and after the Rule, making it possible to estimate the Rule’s impact from comparisons of As concentrations over time. Reductions in As occurrence likely also occurred among non-transient non-community (NTNC) water systems, a smaller class of public water system.

These systems had been excluded from the old 50 $\mu\text{g}/\text{l}$ standard but were required to meet the more stringent 10 $\mu\text{g}/\text{l}$ standard under the Rule. However, NTNC water systems were not required to monitor As prior to the Rule, leaving us little data to identify the Rule’s impact on As occurrence for these systems. This means that our estimates of the causal effects of the Rule on CWS will be partial estimates of the Rule’s total effects on As in drinking water.

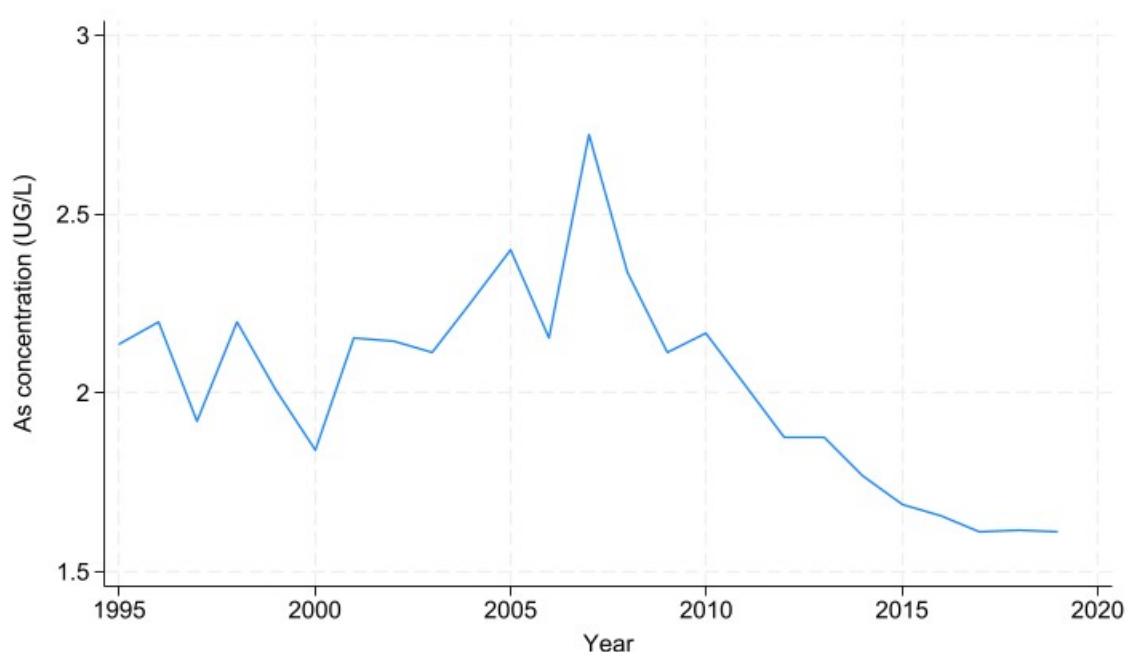
The compliance samples in our data provide actual As concentrations for regulated water systems collected at each entry point to the distribution system, allowing us to create a panel of As sampling records for CWS before and after the Rule. The sampling records also include information on the time and location of the As sample, the sample protocol such as the method detection limit, and details regarding the water system such as county location and whether the CWS uses groundwater or surface water as a source. For most of our records, we also observe whether a sample was collected from an untreated or treated sampling point (i.e., “raw” vs. “finished” water), but we lack this information from 1998 to 2005. We do not exclude raw water samples to avoid introducing inconsistencies in the data over time that could confound effects of the change in the MCL. We select 1995 as the start of our sample period given that fewer systems and states are represented in the data in the 1980s and early 1990s (Figure 6).

We construct a sample that reflects CWS with at least one As sample during the pre-Rule period (1995-2000) and at least one sample during the post-Rule period (2001-2019). We then aggregate the data to the system-by-year level. Collapsing the data to the system-year level yields the mean As concentration for each system i and year t provided the data contain at least one As concentration for system i in year t . Our final sample is an unbalanced system-by-year panel reflecting 22,127 community water systems over 25 years.

Figure 2 depicts annual averages of mean As concentrations for sample CWS over the sample period from 1995 to 2019. From 1995 to the early 2000s, annual averages of mean arsenic concentrations fluctuated around 2 micrograms per liter, but then there is a notable

spike around 2005-2007. After the spike, the figure reveals a downward trend in annual averages of mean As concentrations through the end of the sample period. The increase observed around the 2006 compliance date is likely due to compositional changes. The initial monitoring period under the stricter standard was 2005 to 2007. Ground water systems, which must sample every three years, had to complete sampling by December 31, 2007; the sampling deadline for surface water systems, which are required to sample annually, was December 31, 2006.²⁹ Thus, the spike observed in Figure 2 may have resulted from systems with higher average arsenic concentrations delaying sampling until just prior to the initial sampling deadline. Additionally, systems with higher arsenic concentrations were subject to more frequent testing, which may have further increased the reported average arsenic level during this period. Finally, under the new rule, more water systems were required to report arsenic levels, including those that previously may not have been regularly testing or publicly reporting their results.

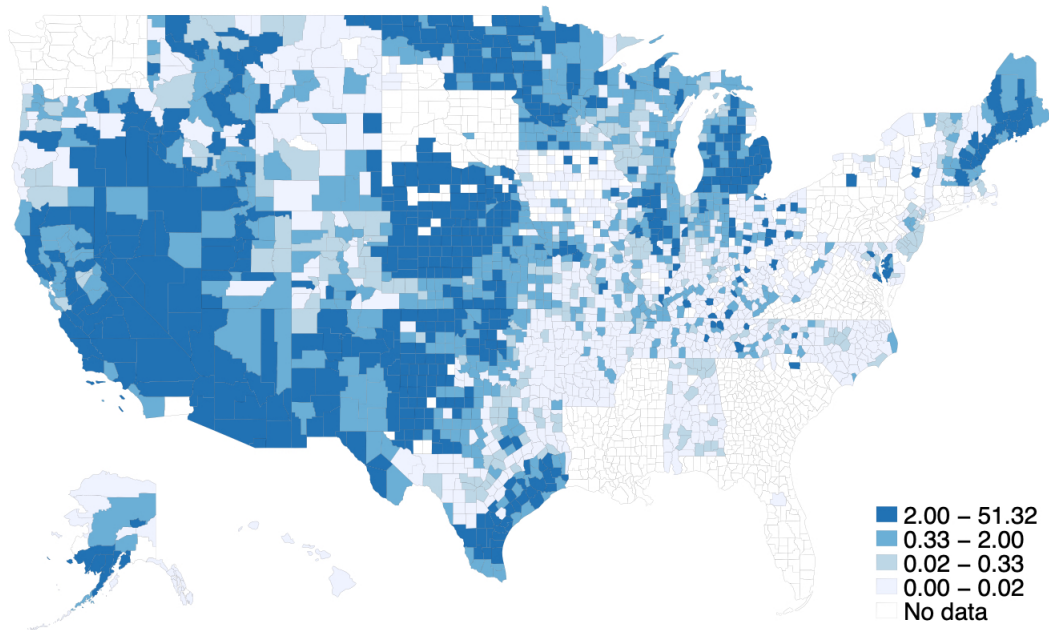
Figure 2: Trends in Average Annual Mean As Concentrations among Sample CWS, 1995 to 2019



To depict spatial variation in baseline As concentrations, we collapse the data at the

county level. Figure 3 shows the average annual mean As concentration by county over the period 1995-2000, prior to the Rule. Systems for which we do not observe a geographic location, about 13.5% of the sample, are not reflected in the figure. Counties in the southeastern US that are represented in our sample generally had lower As concentrations compared to other areas prior to the Arsenic Rule. As concentrations were generally larger in the West and in portions of the Northeast in the pre-Rule period.

Figure 3: Average Annual Mean As Concentration by County, 1995-2000



Obtaining causal effects of the Rule on As concentrations requires a research design that removes concurrent trends in As concentrations (e.g., due to changes in the natural environment, changes in anthropogenic releases, or economic activity) from the estimated impact of the policy. To do so, we employ a difference-in-differences (DD) strategy that defines treatment based on the 2001 Rule’s regulatory threshold of $10 \mu\text{g/l}$. Specifically, we consider treated water systems as those where measured As levels were ever above the proposed regulatory threshold *before* the Rule was published in 2001. Control systems are those for which we never observe an As concentration above $10 \mu\text{g/l}$ prior to 2001. We compare how As concentrations changed for the treated group relative to the control group

before and after the Rule. Because the Rule is not binding for water systems with baseline As levels below the regulatory threshold (i.e., control systems), changes in As in these systems likely reflect correlated changes in As from the environment or economic activity rather than changes due to any mitigation measures.

Almost 9% of systems in our sample, specifically 1952 systems, are assigned to treatment with the remaining systems serving as controls. Table 3 compares average treatment and control systems based on observable system characteristics in the pretreatment period (i.e., prior to 2001). Compared to control systems, treatment systems are on average more likely to be very large (serving more than 100,000 people) or large (serving between 10,001 and 100,000 people) but less likely to be very small (serving fewer than 500 people). In addition, the average treatment system is more likely than the average control system to exclusively source ground water. Finally, relative to control systems on average treatment systems are less likely to be located in the South and Midwest and more likely to be located in the West. These level mean differences in observable system characteristics do not threaten a causal interpretation of our results. However, a simple comparison of raw mean As concentrations for treatment and control systems before and after the Rule’s promulgation fails to control for potential confounding factors so we estimate the following DD regression specification in equation (1).

$$Y_{it} = \beta_0 + \beta_1 \text{aboveMCL}_{it}^{2001} + \gamma_i + \gamma_t + u_{it} \quad (1)$$

Y_{it} represents the mean As concentration for system i in year t . The variable $\text{aboveMCL}_{it}^{2001}$ is equal to 1 for treatment systems in post-treatment years (e.g., after 2000) and 0 otherwise. γ_i and γ_t respectively represent water system and year fixed effects. γ_i controls for time-invariant water system characteristics (e.g., geographic location), and γ_t controls for time-varying factors common to all water systems (e.g., national macroeconomic trends). β_1 represents the DD estimate, the coefficient of interest, which measures how As concentrations for treated systems changed on average relative to control systems following treatment. Under the assumption that the set of control systems reflect how As concentrations would

have changed for the treated systems in absence of the Rule (i.e., parallel trends), β_1 is an estimate of the causal impact of the Rule on CWS.

We also estimate an event-study specification, as in equation (2), to illustrate the dynamic effects of the Rule on As occurrence:

$$Y_{it} = \beta_0 + \sum_{t \in (1995, 2019), t \neq 2000} \alpha_t \cdot treated_i \cdot D_{it} + \gamma_i + \gamma_t + u_{it} \quad (2)$$

In equation (2), $treated_i$, which takes the value of one if system i is a treatment system and zero otherwise, is interacted with a set of indicator variables denoting event time. We express event time in calendar years with 2000, the year prior to the final As rule’s promulgation, as the reference year. The coefficients of interest are the estimates of α_t , which reflects the average difference in mean annual As occurrence between treatment and control systems in year t compared to 2000. Under the assumption of parallel trends, the estimated α_t coefficients in the post-treatment period capture the causal effects of the Rule on As occurrence as they unfold over time. In the pre-treatment period (i.e., before 2001), the α_t are essentially placebo tests, such that statistically non-zero estimates of α_t prior to the Rule would threaten the assumption of parallel trends in As occurrence between control and treatment CWS absent the Rule. The remaining parameters in equation (2) are identical to those in equation (1).

Results and discussion

Differences in means

Table 1 provides averages of annual mean As concentrations for treatment and control systems before and after the Rule’s promulgation in 2001. The averages of annual mean As concentrations for control systems before and after 2001 were respectively 0.744 and 0.991 $\mu\text{g/l}$, well below the 10 $\mu\text{g/l}$ standard. In contrast, the average of annual mean As concen-

trations for treatment systems fell from 12.961 $\mu\text{g/l}$ before 2001 to 8.819 $\mu\text{g/l}$ after the Rule was passed. Viewed differently, annual mean As concentrations for treatment systems were, on average, 12.217 $\mu\text{g/l}$ higher than control systems before 2001, but only 7.829 $\mu\text{g/l}$ higher, on average, after the Rule. These preliminary differences in raw means suggest that the Rule could have decreased annual mean As concentrations for treatment systems by 4.388 $\mu\text{g/l}$ or about 34% relative to their baseline, pre-2001 As levels.

Table 1: Averages of Annual Mean As Concentrations

	Treatment (1)	Control (2)	Difference (1)-(2)
Before 2001	12.961 (17.214)	0.744 (1.692)	12.217 [0.000]
After 2001	8.819 (13.247)	0.991 (3.070)	7.829 [0.000]

^a Note: The table reports means of annual average arsenic (As) concentrations in sample community water systems (CWS) before and after 2001 (inclusive of 2001) by treatment status. Standard deviations are in parentheses. The final column reports the difference in means by treatment status and, in brackets, the p-value associated with a t-test for equality of means. 1952 unique CWS are assigned to the treatment group. 20,175 unique CWS are assigned to the control group.

Table 1 indicates that control systems experienced a small increase in average annual mean As concentrations subsequent to the Rule. This pattern raises concerns that control and treatment groups may have followed different trends even in the absence of the Rule, which would violate the parallel trends assumption required for a causal interpretation of our results. To explore this formally, we assess the robustness of our findings to deviations from the strict parallel trends assumption following Rambachan and Roth³⁰. We discuss these results in the discussion section.

Difference-in-difference estimates

The first column of Table 2 presents the DD estimate of the Rule’s impact (β_1 from equation 1). Standard errors are clustered at the water system level and are provided in parentheses. After 2000, annual mean As concentrations in treatment systems fell by 4.956 $\mu\text{g/l}$ ($p < 0.01$) relative to control systems, representing a decrease of about 38% compared to pre-2001 levels. In the second column of Table 2, we show robustness to an alternative definition of the post-

treatment period: 1997 and later, after the impending rulemaking had been announced but prior to resolution of uncertainty regarding specifics of the Rule. Between 1997 and 2001, some systems may have invested in control technologies in anticipation of the impending Rule. The results reported in column (2) explore this. Using the alternative treatment definition (i.e., 1997+), the DD estimate suggests an average reduction of 3.576 $\mu\text{g}/\text{l}$ in annual mean As concentrations among treatment systems compared to control systems, slightly lower than our main DD estimate. Results are qualitatively similar when we estimate a log-linear version of equation 1 (Table 4).

Table 2: Estimated Impact of the Rule on Annual Mean As Concentrations

	Post-treatment period	
	2001+ (1)	1997+ (2)
β_1	-4.956*** (0.303)	-3.576*** (0.336)
Pre-treatment average	12.961	12.213
Estimated effect size	-38.240%	-29.04%

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

^a Notes: The table presents the estimated impact of the Rule on annual mean arsenic (As) concentrations during two post-treatment periods: 2001+ and 1997+. Standard errors are clustered at the water system level and are provided in parentheses. Statistical significance is denoted as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sample size is 192,347 observations, representing 22,127 unique community water systems. Pre-treatment average is the average of annual mean As concentration for treatment systems in the pre-treatment period, as defined in the respective column.

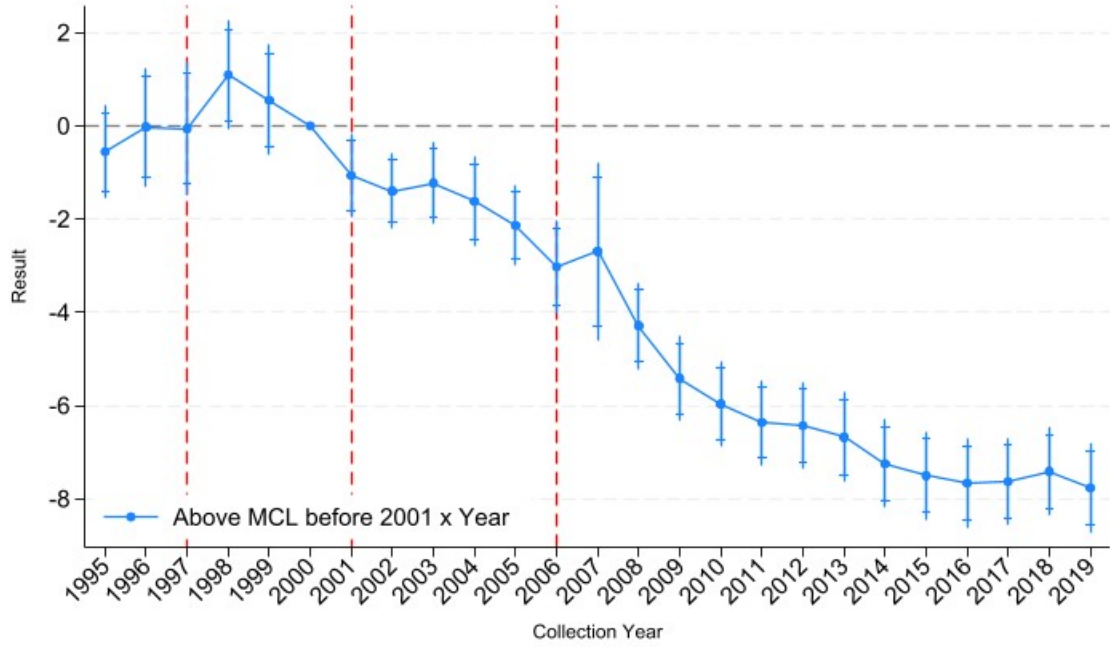
How does the size of the estimated reduction in As based on our retrospective analysis compare to the results of the prospective Regulatory Impact Analysis (RIA) conducted by EPA? When projecting the likely benefits of the Rule in its *ex ante* RIA, EPA assumed that affected systems would reduce As concentrations to 0.8 times the new 10 $\mu\text{g}/\text{l}$ standard (or to 8 $\mu\text{g}/\text{l}$) on average.³¹ To produce a post-Rule As concentration based on our retrospective analysis that is appropriate for comparison, we subtract an estimated value of β_1 in Table 2 from the relevant pre-treatment average. Using the 2001 post-treatment period definition, the estimated post-Rule average annual mean As concentration for treated CWS based on our DD estimate is 8.005 $\mu\text{g}/\text{l}$, remarkably similar to the prospective RIA's estimate. We

obtain a slightly higher estimate of 8.637 $\mu\text{g/l}$ when we use the alternative post-treatment period definition (i.e., 1997+).

Event study estimates

Figure 4 reports results from estimating the event study model in equation (2), where we recover an estimated impact of the Rule for each post-treatment year relative to 2000. The estimated coefficient for year t reflects the average difference in mean As occurrence between treatment and control systems in year t compared to 2000. Beginning in 2001, mean As concentrations among treatment systems show a statistically significant decrease compared to control systems, relative to the base year. These declines continue through the early 2010s when they start to level off. Figure 4 also shows limited evidence of differential trends in mean As concentrations between treatment and control systems before the Rule was passed, which strengthens the interpretation of our DD and event study results as reflecting the estimated *causal* change in As from the Rule. As demonstrated in the Supplemental Information, results are similar with a log-linear specification (Figure 7), when we restrict our analysis to the states included in the original occurrence data used to support the Rule (Figure 8), and when we use 1997 instead of 2001 as the base year for comparison (Figure 9).

Figure 4: Event Study Estimates



^a Notes: The figure shows estimated coefficients and 95% confidence intervals from an event study specification with annual mean arsenic (As) concentrations as the dependent variable. The base year for comparison is 2000.

Variation in effects related to system characteristics

The difference-in-differences and event study results provide estimates of the average effect of the Rule on mean As concentrations for community water systems. While an estimate of the average effect is useful, if different types of CWS face different costs of compliance with the stricter MCL, then we might expect to see variation in the Rule's effects across systems. Indeed, EPA's prospective RIA of the Rule estimated variation in compliance costs across systems based on treatment technology, system size, and source water, among other factors.^{31,32} We explore three different dimensions of heterogeneity based on observable system characteristics: system size (i.e., by service population), region, and water source. It is important to note, however, that these system characteristics are not independent. For example, very small (serving fewer than 500 people), small (serving between 501 and 3,300 people), and medium (serving 3,301 and 10,000 people) systems are more likely to use groundwater sources exclusively compared to large (serving 10,001 and 100,000 people) and

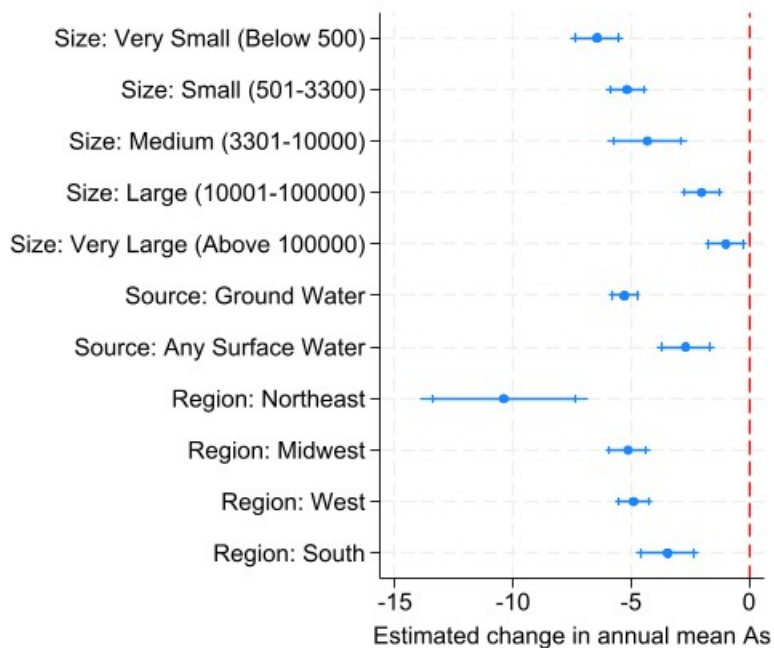
very large systems (serving more than 100,000 people), which are more likely to have some surface water sources.

Figure 5 reports our results from the tests for heterogeneous effects. Each estimate in the figure is from a different regression in which we estimate β_1 from equation (1) for different sub-samples defined based on observed system characteristics. Figure 5 reports these estimates along with 95% confidence intervals.

Consider, first, the results with respect to system size. Consistent with the *ex ante* cost estimates in EPA’s prospective RIA, the estimated reductions in As concentrations from the Rule occur primarily in smaller CWS, especially those serving fewer than 10,000 people. The estimated β_1 for very small systems is $-6.445 \mu\text{g/l}$, corresponding to about a 38% reduction when evaluated at the pre-treatment subsample mean for very small treatment systems of $16.756 \mu\text{g/l}$. In contrast, the estimated β_1 for large systems is $-2.015 \mu\text{g/l}$, about a 29% reduction when evaluated at the pre-treatment subsample mean for large treatment systems of $6.829 \mu\text{g/l}$. That smaller water systems, which are more likely to rely on groundwater sources, saw the largest impacts of the Rule is consistent with concerns over high compliance costs for small systems. Note, however, that the larger estimated reduction for small systems is also suggestive of larger benefits for communities served by smaller systems.

We also estimate different effects of the Rule based on source water. The estimated reduction for ground water-sourced systems is almost twice that of systems that source some surface water. Our results for system size and source water are consistent with findings from Foster et al.¹⁶ that most of the increased violations following the Rule occurred in small systems and those using groundwater. Lastly, we explore regional variation in the Rule’s estimated impact. Figure 5 shows similar estimated effects for systems located in the Midwest and West. Compared to these regions, estimated reductions are slightly smaller for systems in the South and largest for systems in the Northeast.

Figure 5: Heterogeneous Effects by System Size, Water Source and Region



^a Notes: The figure reports estimates of β_1 and associated 95% confidence intervals from regressions of the form in equation (1) with subsamples based on observed CWS characteristics. Each estimate in the figure comes from a different subsample regression reflecting the restriction described in the first column.

Discussion

Taken together, our retrospective analysis suggests that the 2001 Arsenic Rule under the SDWA significantly reduced As occurrence among U.S. community water systems. Statistically significant decreases in average As occurrence begin as early as 2001 and continue through the last year of our sample, 2019. Reductions in As are largest for small systems and those relying exclusively on groundwater as a source, and they vary significantly by region.

As noted above, our approach is designed to identify the causal effect of the As Rule on As occurrence among U.S. community water systems. Our models compare differences pre- vs. post-Rule between systems for which the Rule was likely to bind (i.e., those that had exceeded the new MCL prior to its promulgation) and a set of control systems that had never exceeded the new MCL pre-policy. Remaining sources of confounding variation would have to affect only treated systems, and only post-policy, in order to materially affect

our results. One concern might be that treatment CWS could have adopted new treatment technologies for compliance with other rules passed under the SDWA around the time of the Arsenic Rule, and that those technologies could also affect As occurrence. However, most of the other SDWA rules were enacted well before the effective date of the Arsenic Rule, so they are unlikely to have contributed to the reductions in arsenic levels observed in our analysis. For example, activated alumina, a treatment used to remove fluoride, also absorbs arsenic³³. However, the Fluoride Rule was passed in 1986 so installation of treatment technologies or processes in response likely began well before the early 2000s, when our data first show a decline in arsenic. To our knowledge, other rules promulgated around the time of the Arsenic Rule would not cause systems to adopt technologies or processes that would reduce As.

Bias could also result from the fact that the compliance samples in our data are self-reported by CWS, and the literature identifies strategic sampling as a challenge for monitoring and enforcement under some SDWA rules, particularly those in which violations occur once the number of problematic samples exceeds a percentage threshold.^{34,35} Some aspects of the As Rule use percentage thresholds to determine compliance, and if CWS are sampling strategically under the Rule, such results could be included in our data. If this behavior increased for treated systems starting in 2001 and it resulted in additional lower-than-MCL samples for treated systems, our approach could over-estimate the Rule’s impact on As occurrence.

An additional threat to identification arises from increased testing requirements under the Arsenic Rule, which mandate that CWS exceeding the MCL must test more frequently and publicly report results. This increased scrutiny and testing frequency could lead to the detection of higher arsenic levels in systems we initially classified as controls. Such factors could cause violations of the parallel trends assumption required for a causal interpretation of our DD estimates. To assess the implications of potential violations of parallel trends, we apply the “honest” difference-in-differences (HonestDiD) approach proposed by Rambachan and Roth³⁰ to our event study estimates. Figure 10 presents the results. In the figure, $\bar{M}$

represents the magnitude of deviations from the parallel trends assumption permitted post-treatment. We applied the HonestDiD test separately to each post-treatment event study estimate from 2001 to 2019. We find that estimated treatment effects for 2001–2005, during the early Rule implementation period, are sensitive to potential violations of the parallel trends. However, from about 2006 onward, the estimates become increasingly robust to such violations. Between 2009 and 2019, the estimated treatment effects remain statistically significant and of similar magnitude even when allowing for large post-treatment deviations up to $\bar{M} = 2$, meaning treatment and control group trends could diverge by up to 200% of the maximum deviation observed in the pre-treatment period. These findings suggest that our results for the later period, after the Rule’s full implementation, are robust to substantial violations of the parallel trends assumption.

A final potential limitation of our analysis is our inability to distinguish between “raw” and “finished” water samples for part of the sample period (i.e., 1998 to 2005), leading us to retain raw water samples for all years. Raw water tends to have higher levels of contaminants than treated water, and hence this data selection decision could result in higher levels of arsenic than were actually present in tap water. To test the sensitivity of our results to this concern, we drop all identifiable raw water samples and re-estimate the main DD specification using this subsample. Estimated DD coefficients are similar, albeit slightly larger in magnitude, when we drop raw water samples (e.g., -5.4 compared to -5.0 assuming a treatment year of 2001).

Our causal results for As occurrence are suggestive of health benefits from the As Rule, though we leave the quantification of health benefits for future work. Moreover, a retrospective analysis of the Rule’s costs would allow for a re-evaluation of the EPA’s prospective conclusion in its regulatory impact analysis that the Rule’s benefits were unlikely to exceed its costs, particularly for small systems. The SDWA six-year review data that we use to estimate the Rule’s effect on As occurrence also include information on the drinking water treatment technologies in place at CWS. Unlike the analytic sampling data, the EPA has

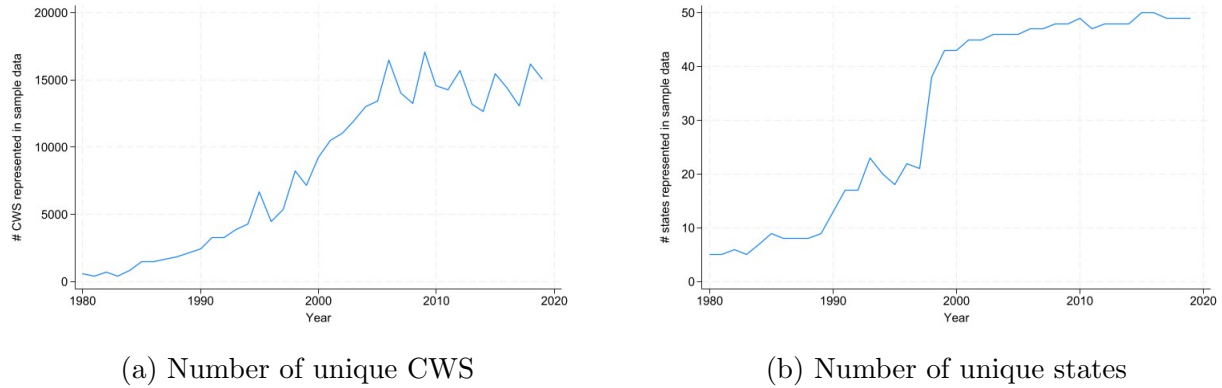
not applied quality assurance evaluations to these treatment technology data in the six-year reviews.²² While the existing literature suggests that the capital costs of compliance with the As Rule may have been lower than anticipated in the RIA,³² determining whether updated compliance cost estimates from the six-year review or other sources can be used to comprehensively update EPA’s prospective cost estimate is an important area for future research.

Acknowledgement

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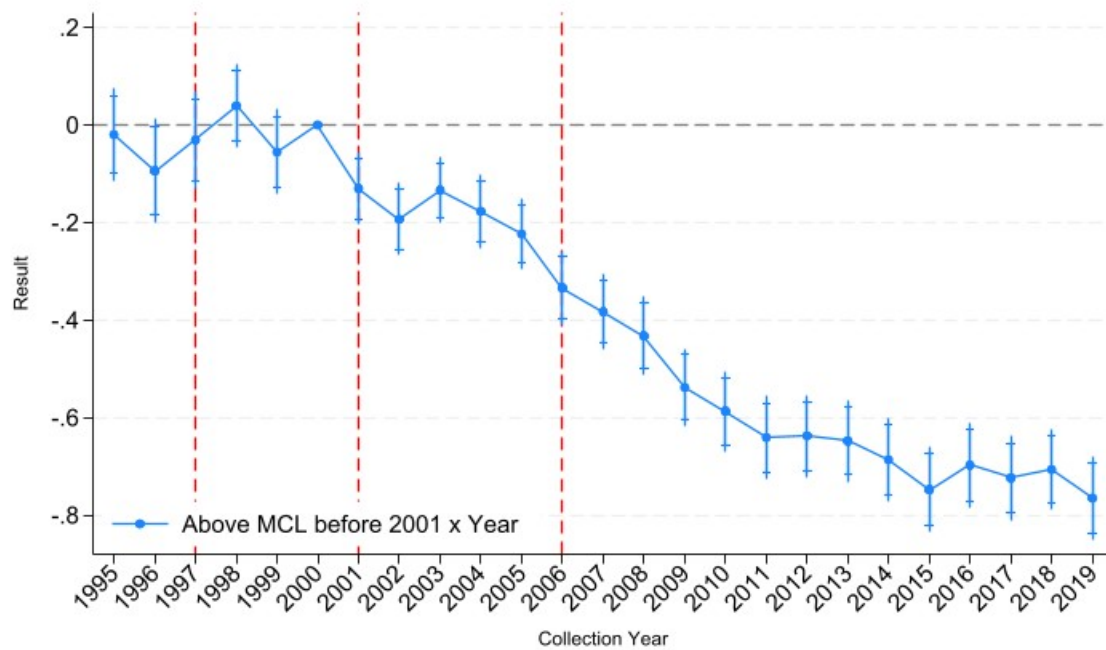
Supporting Information Available

Figure 6: Data Coverage of Systems and States Over Time



^a Notes: Panels (a) and (b) respectively show the number of unique community water systems (CWS) and unique states represented in the data for each year from 1980 to 2019.

Figure 7: Event Study Estimates with Log Specification



^a Notes: Figure shows estimated coefficients and 95% confidence intervals from an event study specification with log-transform annual mean arsenic (As) concentrations as the dependent variable. The base year for comparison is 2000.

Table 3: Balance Table

	Treatment Mean/(SD) (1)	Control Mean/(SD) (2)	Pairwise t-test Mean difference (1)-(2)
Size: Very Small (Below 500)	0.533 (0.499)	0.565 (0.496)	-0.032*** [0.007]
Size: Small (501-3300)	0.253 (0.435)	0.264 (0.441)	-0.012 [0.267]
Size: Medium (3301-10000)	0.101 (0.301)	0.094 (0.292)	0.007 [0.317]
Size: Large (10001-100000)	0.094 (0.292)	0.070 (0.255)	0.024*** [0.000]
Size: Very Large (Above 100000)	0.019 (0.138)	0.007 (0.086)	0.012*** [0.000]
Source: Ground Water	0.924 (0.265)	0.869 (0.337)	0.055*** [0.000]
Region: Northeast	0.082 (0.275)	0.091 (0.287)	-0.008 [0.223]
Region: South	0.220 (0.415)	0.373 (0.484)	-0.152*** [0.000]
Region: Midwest	0.221 (0.415)	0.308 (0.462)	-0.086*** [0.000]
Region: West	0.476 (0.500)	0.229 (0.420)	0.247*** [0.000]

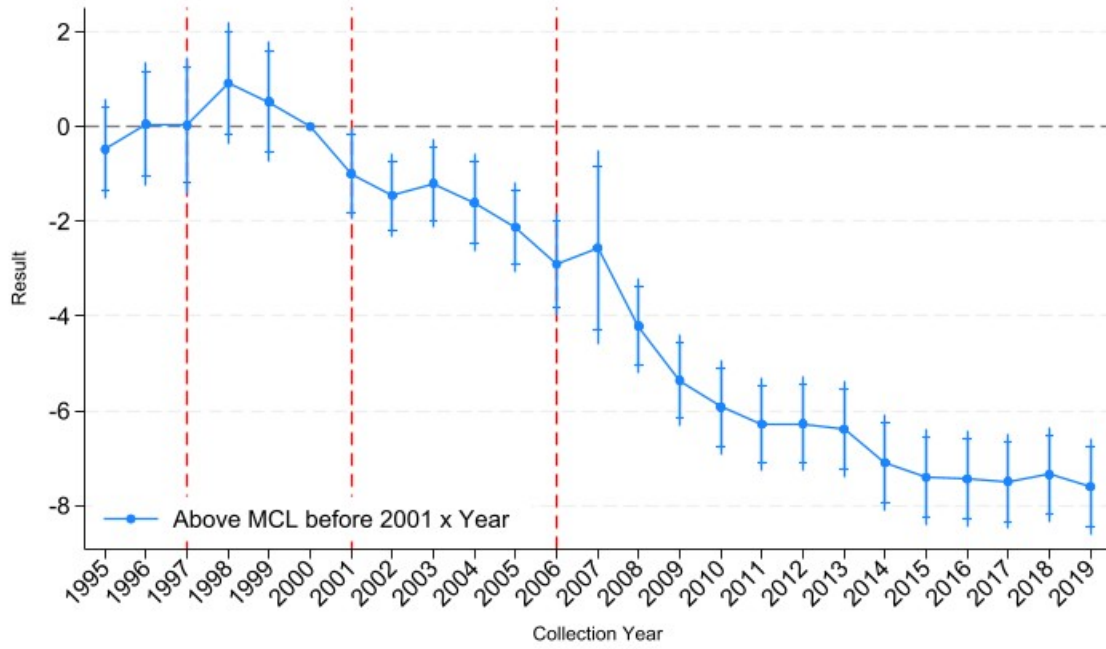
^a Notes: The table reports the means of community water system (CWS) characteristics by treatment status. Standard deviations are in parentheses. The final column reports the difference in means by treatment status and, in brackets, the p-value associated with a t-test for equality of means. 1,952 unique CWS are assigned to the treatment group. 20,175 unique CWS are assigned to the control group. Statistical significance is denoted as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 4: Estimated Impact of the Arsenic Rule on Log Mean As Concentrations

	Post-treatment period	
	2001+ (1)	1997+ (2)
β_1	-0.452*** (0.0208)	-0.354*** (0.0307)
Estimated change in As concentration	-4.715	-3.669
Pre-treatment average	12.961	12.312
Effect size	-36.382%	-29.800%

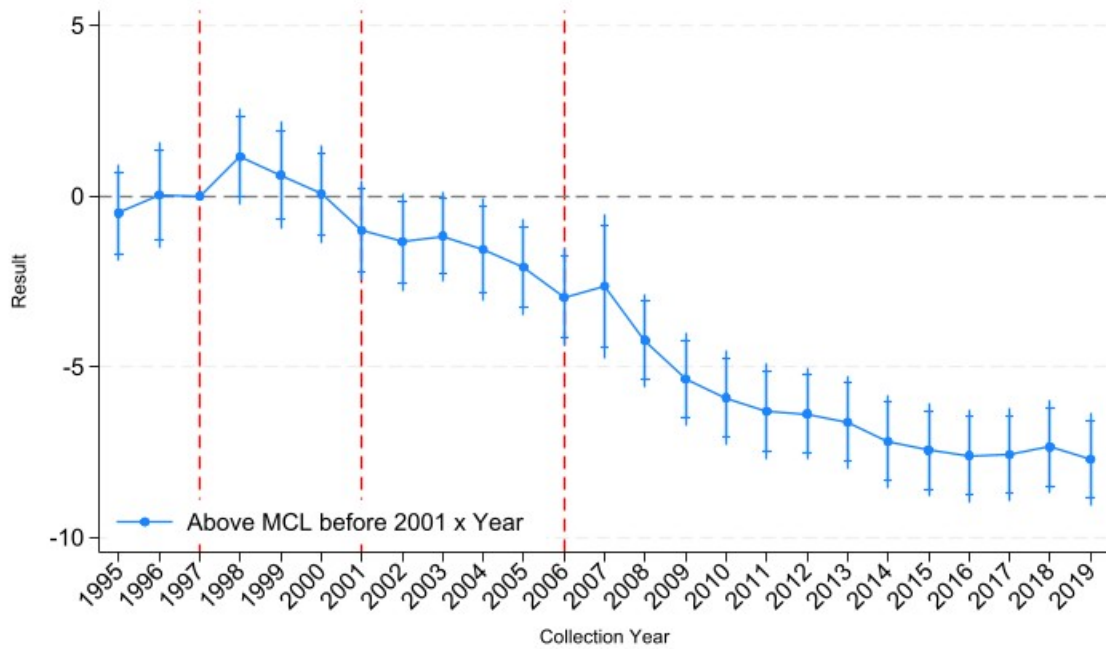
^a Notes: The table presents the estimated impact of the Rule on log-transformed annual mean arsenic (As) concentrations during two post-treatment periods: 2001+ and 1997+. Standard errors are clustered at the water system level and are provided in parentheses. Estimated changes in annual mean As concentrations are calculated by exponentiating the coefficients and applying them to the pre-treatment average of mean As concentrations for the treatment group. Effect sizes are reported as percentage changes relative to the pre-treatment average. Pre-treatment average is the average of annual mean As concentration for treatment systems in the pre-treatment period, as defined in the respective column. Sample size is 64,624 observations, representing 10,127 unique community water systems. Statistical significance is denoted as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure 8: Event Study Results with Subsample of 24 States



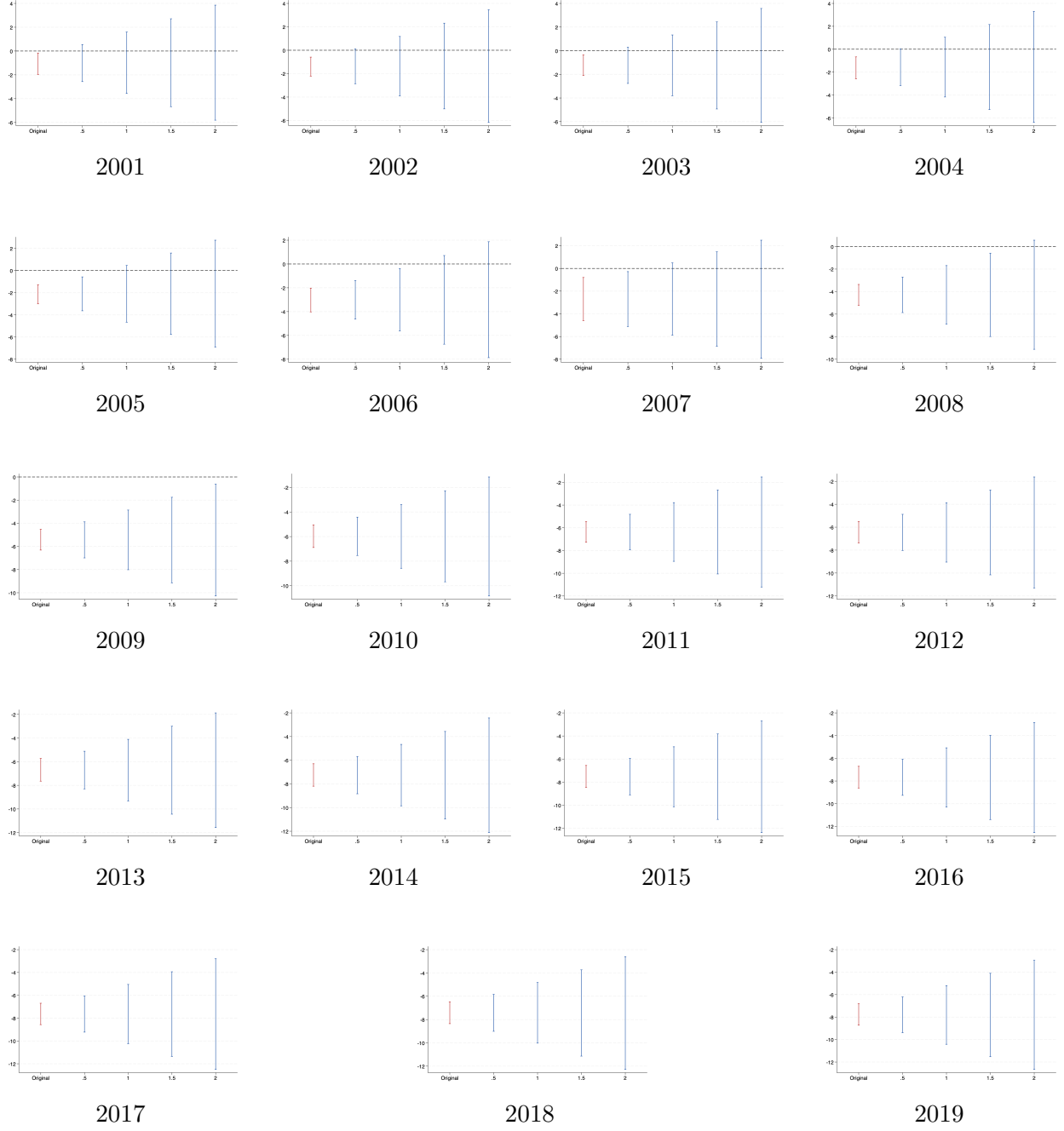
^a Notes: Figure shows estimated coefficients and 95% confidence intervals from an event study specification with annual mean arsenic (As) concentrations as the dependent variable. The sample is restricted to CWS located in the 24 states represented in the Rule occurrence data. The base year for comparison is 2000.

Figure 9: Event Study Results with Alternative Post-Treatment Period



^a Notes: Figure shows estimated coefficients and 95% confidence intervals from an event study specification with annual mean arsenic (As) concentrations as the dependent variable. The base year for comparison is 1997.

Figure 10: HonestDiD Results for Event Study Estimates from 2001 to 2019.



^a Notes: Figure shows the HonestDiD sensitivity analyses for each post-treatment year from 2001 to 2019, using event study estimates. The base year for comparison is 2000. Higher M values on the horizontal axis further relax the parallel trends assumption by allowing larger deviations between treatment and control systems post-treatment.

References

- (1) EPA *IRIS Toxicological Review of Inorganic Arsenic*; 2025; Accessed: 2025-01-28.
- (2) Bedaiwi, A.; Wysong, A.; Rogan, E. G.; Clarey, D.; Arcari, C. M. Arsenic exposure and melanoma among US adults aged 20 or older, 2003-2016. *Public Health Reports* **2022**, *137*, 548–556.
- (3) Issanov, A.; Adewusi, B.; Dummer, T. J.; Saint-Jacques, N. Arsenic in Drinking Water and Urinary Tract Cancers: A Systematic Review Update. *Water* **2023**, *15*, 2185.
- (4) Shao, K.; Zhou, Z.; Xun, P.; Cohen, S. M. Bayesian benchmark dose analysis for inorganic arsenic in drinking water associated with bladder and lung cancer using epidemiological data. *Toxicology* **2021**, *455*, 152752.
- (5) Medgyesi, D. N.; Bangia, K.; Spielfogel, E. S.; Fisher, J. A.; Madrigal, J. M.; Jones, R. R.; Ward, M. H.; Jr., J. V. L.; Sanchez, T. R. Long-term exposure to arsenic in community water supplies and risk of cardiovascular disease among women in the California Teachers Study. *Environmental Health Perspectives* **2024**, *132*, 107006.
- (6) Yang, K.; Xun, P.; Carnethon, M.; Carson, A. P.; Lu, L.; Zhu, J.; He, K. Low to moderate toenail arsenic levels in young adulthood and incidence of diabetes later in life: findings from the CARDIA Trace Element study. *Environmental research* **2019**, *171*, 321–327.
- (7) Bloom, M. S.; Surdu, S.; Neamtiu, I. A.; Gurzau, E. S. Maternal arsenic exposure and birth outcomes: a comprehensive review of the epidemiologic literature focused on drinking water. *International journal of hygiene and environmental health* **2014**, *217*, 709–719.
- (8) Tsuji, J. S.; Garry, M. R.; Perez, V.; Chang, E. T. Low-level arsenic exposure and devel-

- opmental neurotoxicity in children: A systematic review and risk assessment. *Toxicology* **2015**, *337*, 91–107.
- (9) Service, C. R. *Arsenic in Drinking Water: Regulatory Developments and Issues*; 2007; Accessed: 2025-01-28.
- (10) Burnett, J. K.; Hahn, R. W. A Costly Benefit: Economic Analysis Does Not Support EPA’s New Arsenic Rule. *Regulation* **2001**, *24*, 44.
- (11) Belzer, R. B. Achieving Economically Feasible Drinking Water Regulation. *Journal of Benefit-Cost Analysis* **2020**, *11*, 294–318.
- (12) Jones, S. A.; Joy, N. The inequity of the Revised Arsenic Rule for very small community drinking water systems. *Environmental Science & Policy* **2006**, *9*, 555–562.
- (13) Greenstone, M. Toward a culture of persistent regulatory experimentation and evaluation. *New perspectives on regulation* **2009**, *111*, 116–19.
- (14) Morgenstern, R. Retrospective Analysis of US Federal Environmental Regulation. *Journal of Benefit-Cost Analysis* **2018**, *9*, 285–304.
- (15) Cropper, M. L.; Morgenstern, R. D.; Rivers, N. Policy Brief—Facilitating Retrospective Analysis of Environmental Regulations. *Review of Environmental Economics and Policy* **2018**,
- (16) Foster, S. A.; Pennino, M. J.; Compton, J. E.; Leibowitz, S. G.; Kile, M. L. Arsenic drinking water violations decreased across the United States following revision of the maximum contaminant level. *Environmental science & technology* **2019**, *53*, 11478–11485.
- (17) Grooms, K. K. Does Water Quality Improve When a Safe Drinking Water Act Violation Is Issued? A Study of the Effectiveness of the SDWA in California. *The B.E. Journal of Economic Analysis & Policy* **2016**, *16*, 1–23.

- (18) Grooms, K.; Ferrazares, T.; Royer, H.; Schnepel, K. Drinking Water and Infant Health: Evidence from Contaminant Levels in California. *Working Paper* **2023**,
- (19) Nigra, A. E.; Sanchez, T. R.; Nachman, K. E.; Harvey, D. E.; Chillrud, S. N.; Graziano, J. H.; Navas-Acien, A. The effect of the Environmental Protection Agency maximum contaminant level on arsenic exposure in the USA from 2003 to 2014: an analysis of the National Health and Nutrition Examination Survey (NHANES). *The lancet public health* **2017**, *2*, e513–e521.
- (20) EPA *Arsenic Occurrence in Public Drinking Water Supplies*; 2000; Accessed: 2022-02-10.
- (21) Agency, U. E. P. *The Analysis of Regulated Contaminant Occurrence Data from Public Water Systems in Support of the Second Six-Year Review of National Primary Drinking Water Regulations*; 2010; Accessed: 2025-01-28.
- (22) EPA *The Data Management and Quality Assurance Quality Control Process for the Third Six-Year Review Information Collection Rule Dataset*; 2016; Accessed: 2022-02-10.
- (23) Agency, U. E. P. *EPA Protocol for the Fourth Review of Existing National Primary Drinking Water Regulations*; 2024; Accessed: 2025-01-28.
- (24) Zheng, J. *Lead exposure, human capital formation, and inequality: the impacts of drinking water lead on short-run and long-run educational outcomes*; 2021.
- (25) Keiser, D. A.; Molitor, D.; Shapiro, J. S. Water works: Causes and consequences of safe drinking water in America. 2023.
- (26) Austin, W.; Pan, S.; Parthum, B. M. When it Rains, it Pours: Severe Weather Events, Flooding, and Drinking Water Quality. **2025**,

- (27) Information about Public Water Systems. <https://www.epa.gov/dwreginfo/information-about-public-water-systems>, Accessed: 2025-03-05.
- (28) Community Water System Service Area Boundaries. <https://www.epa.gov/ground-water-and-drinking-water/community-water-system-service-area-boundaries>, Accessed: 2025-03-05.
- (29) The Arsenic Rule: Compliance, Reporting, and Enforcement Issues. <https://www.epa.gov/sites/default/files/2015-09/documents/train4-compliance.pdf>, Accessed: 2025-03-05.
- (30) Rambachan, A.; Roth, J. A more credible approach to parallel trends. *Review of Economic Studies* **2023**, *90*, 2555–2591.
- (31) Abt Associates, Inc. *Arsenic in Drinking Water Rule Economic Analysis*; 2000; Accessed: 2023-02-26.
- (32) Morgan, C.; Simon, N. National primary drinking water regulation for arsenic: A retrospective assessment of costs. *Journal of Benefit-Cost Analysis* **2014**, *5*, 259–284.
- (33) EPA Point-of-use or point-of-entry treatment options for small drinking water systems. **2006**,
- (34) Benneer, L. S.; Jessoe, K. K.; Olmstead, S. M. Sampling out: Regulatory avoidance and the total coliform rule. *Environmental Science Technology* **2009**, *43*.
- (35) Andarge, T.; Ghanem, D.; Keiser, D. A.; Lade, G. E. *Quantifying Threshold Manipulation in the Presence of Rounding: The Case of Lead Monitoring in U.S. Drinking Water*; 2024.

TOC Graphic

